

Cassandra_BDA.docx

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Summary

This technical guide provides a practical overview of managing data within **Cassandra**, a distributed NoSQL database, by utilizing the **CQL shell interface**. It outlines the foundational hierarchy of the system, starting with the creation of a **Keyspace**—a container for data—and moving into the definition of **Tables** with specific primary keys. The text details essential **CRUD operations** for manipulating records, while highlighting specialized features like **Secondary Indexes** for non-primary key queries and **Collection types** for storing sets and lists. Finally, it covers advanced utility functions such as **Counters**, **Time To Live (TTL)** for temporary data, and methods for **importing or exporting** datasets via CSV files.

Keywords: Apache Cassandra Commands, Keyspace and Tables, CRUD Operations, Data Collections, Import and Export

Content

Working with CassandraNote: type cqlsh in Ubuntu terminal to start cassandraCreate KeySpace :CREATE KEYSPACE Students WITH REPLICATION = {'class':'SimpleStrategy','replication_factor':1};Describe the existing Keyspaces:DESCRIBE KEYSPACES;For More details on existing keyspaces:SELECT * FROM system.schema_keyspaces;use the keyspace "Students":USE Students;To create table (column family) by name Student_Info:CREATE TABLE Students_Info (Roll_No int PRIMARY KEY, StudName text, DateOfJoining timestamp, last_exam_Percent double);Lookup the names of all tables in the current keyspacesDESCRIBE TABLES;Describe the table informationDESCRIBE TABLE <Table_Name>;CRUDInsert :BEGIN BATCH INSERT INTO Students_Info(Roll_No, StudName, DateOfJoining, last_exam_Percent)VALUES (1,'Asha','2012-03-12',79.9)INSERT INTO Students_Info(Roll_No, StudName, DateOfJoining, last_exam_Percent)VALUES (1,'Krian','2012-03-12',89.9)INSERT INTO Students_Info(Roll_No, StudName, DateOfJoining, last_exam_Percent)VALUES (1,'Tarun','2012-03-12',78.9)INSERT INTO Students_Info(Roll_No, StudName,

DateOfJoining, last_exam_Percent)VALUES (1,'Samrth','2012-03-12',90.9)INSERT
 INTO Students_Info(Roll_No, StudName, DateOfJoining, last_exam_Percent)VALUES
 (1,'Smitha','2012-03-12',67.9)INSERT INTO Students_Info(Roll_No, StudName,
 DateOfJoining, last_exam_Percent)VALUES (1,'Rohan','2012-03-12',56.9)APPLY
 BATCH;View data from the table "Students_Info"SELECT * FROM Students_Info;View
 data from the table "Students_Info" where RollNo column either has a value 1 or 2
 or 3SELECT * FROM Students_Info WHERE Roll_No IN (1,2,3);To execute a non
 primary key - will throw an errorselect * from students_info where Studname=
 'Asha';So create an INDEX on the Column as below:To create an INDEX on StudName
 Column of the Students_Info column familyCREATE INDEX ON Students_Info
 (StudName);Now execute the query based on the INDEXED Column:select * from
 students_info where Studname= 'Asha';To specify the number of rows returned in the
 outputselect Roll_No, StudName from students_info LIMIT 2;Alias for Column:Select
 Roll_No as "USN" from Students_info;UPDATEUPDATE students_info SET
 StudName='David Sheen' WHERE RollNo=2;Lets try to update the primary
 keyUPDATE students_info SET rollno=6 WHERE rollno=3;DELETEDELETE
 LastExamPercent FROM students_info WHERE RollNo=2;Delete a RowDELETE FROM
 student_info WHERE RollNo=2;Set CollectionA column of type set consists of
 unordered unique values. However, when the column is queried, it returns, it returns
 the values in sorted order. For example, for text values, it sorts in alphabetical
 order.ALTER TABLE students_info ADD hobbies set<text>List CollectionWhen the
 order of elements matter, one should go for a list collection.ALTER TABLE
 students_info ADD language list<text>;UPDATE students_info SET
 hobbies=hobbies+{'Chess,Table Tennis'} WHERE RollNo=1;SELECT * from
 students_info WHERE RollNo=1;UPDATE students_info SET languagse=language+
 ['Hindi,English'] WHERE RollNo=1;Note: You can remove an element from a set using
 the subtraction(-) operator.USING A COUNTERA counter is a special column that is
 changed in increments. For example, we may need a counter column to count the
 number of times a particular book is issued from the library bythe student.CREATE
 TABLE library_book(counter_value counter, book_name varchar, stud_name varchar,
 PRIMARY KEY(book_name,stud_name));Load data into the counter columnUPDATE
 library_book SET counetr value=couner_vale+1 WHERE book_name='Big data
 Analytics' AND stud_name='jeet';TIME TO LIVECREATE TABLE userlogin(userid int
 PRIMARY KEY, password text);INSERT INTO userlogin(userid, password) VALUES
 (1,'infy') USING TTL 30;SELECT TTL(password) FROM userlogin WHERE
 userid=1;IMPORT and EXPORTExport to CSVCOPY elearninglists(id,course_order,
 course_id,courseowner,title) TO 'd:.csv';Import from CSVCOPY
 elearninglists(id,course_order, course_id,courseowner,title) FROM 'd:.csv';Import
 FROM STDINCOPY persons(id,fname,lname)FROM STDIN;Export to STDOUTCOPY
 elearninglists(id,course_order, course_id,courseowner,title) TO STDOUT;